Blockchain Applications: Practice Exam

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Instructions. This practice exam covers Blockchain Analytics (Lectures 1–7): 35 multiple-choice questions, each with four options and one correct answer, one point each, no negative marking. Questions 1–20 test concepts and the reading of course output; questions 21–35 are built on three short cases, each followed by five questions. No calculation beyond one-step mental arithmetic is needed. The real exam allows no books, notes or other materials, lasts 60 minutes and has 30 questions: 15 on Crypto Economics and 15 on Blockchain Analytics, including one case with 5 questions.

All numbers come from the lectures (EOD Historical Data, EODHD, prices to 23 October 2026). Abbreviations: VaR Value-at-Risk (reported as a positive loss, VaR $= -q$), ES expected shortfall, GARCH generalised autoregressive conditional heteroskedasticity, IGARCH integrated GARCH, HS historical simulation, EWMA exponentially weighted moving average, bp basis point (0.01%), HHI Herfindahl–Hirschman index, AUC area under the receiver operating characteristic curve, CAR cumulative abnormal return, DSR deflated Sharpe ratio, QS quantile score, DM Diebold–Mariano.

1. The estimate $\hat{H} = D \cdot 2^{32}/600$, with D the Bitcoin mining difficulty, measures (a) transactions per second; (b) hashes computed per second to secure the network; (c) the number of active miners; (d) the growth rate of the money supply.

2. The Herfindahl–Hirschman index of the five largest coins is 0.71. This means (a) 71% of coins are stablecoins; (b) the market is close to perfect competition; (c) market value is highly concentrated in a few coins; (d) the five coins have correlation 0.71.

3. After the 2024 halving, Bitcoin issuance inflation is about 0.82% a year. This follows from (a) a fixed supply rule: the block reward halves every 210 000 blocks, towards 21 million BTC; (b) a decision of the miners each year; (c) the hash rate; (d) the price of Bitcoin.

4. The Jarque–Bera statistic for daily Bitcoin returns is 29 620. Therefore (a) returns are normally distributed; (b) the variance is infinite; (c) returns are independent; (d) normality is rejected, and a VaR based on the Normal distribution understates crash risk.

5. Ljung–Box $Q(10)$ is 212.7 for squared returns r_t^2 and 20.2 for returns r_t. This shows (a) returns can be predicted profitably; (b) volatility clustering is much stronger than any autocorrelation of returns; (c) returns are independent; (d) returns are normal.

6. Robust variance-ratio tests do not reject weak-form efficiency in any sub-period. The correct conclusion is (a) prices are proven to follow a random walk; (b) there are no bubbles; (c) the data give no evidence of return predictability at the tested horizons; (d) returns are normal.

7. A log-periodic power law fits the 2017 Bitcoin bubble with $R^2 \approx 0.98$. Hence (a) a good in-sample fit does not give a reliable forecast of the crash date; (b) the crash date is known precisely; (c) the model is rejected; (d) the bubble did not exist.

8. For Bitcoin, GARCH(1,1)-t gives $\hat{\alpha} + \hat{\beta} = 1.000$. This means (a) volatility does not cluster; (b) volatility shocks do not die out in this sample (integrated GARCH); (c) a volatility shock halves in 36 days; (d) returns are independent.

9. A 1% VaR model has 45 breaches where 24.5 are expected; the Kupiec test has $p < 0.001$. The model (a) is too conservative; (b) passes the backtest; (c) underestimates risk; (d) has dependent breaches.

10. With Normal-based expected shortfall, realised Bitcoin losses beyond VaR are 26–31% larger than predicted. This means (a) the VaR level is wrong; (b) ES is always larger than VaR; (c) the Normal model is too conservative; (d) the Normal distribution understates how deep losses go once the VaR is breached.

11. During the SVB weekend USDC fell to −1 036 bp, while the value of a token with zero recovery on the SVB deposit was −825 bp. The extra discount reflects (a) a liquidity premium: redemption was impossible while banks were closed; (b) the expected credit loss; (c) a data error; (d) a change of the peg.

12. USDC has the smallest 1% quantile of USDT, USDC and DAI (−7.6 bp) but the largest kurtosis (1 045). Together these numbers say (a) USDC is the safest coin in every respect; (b) USDC is usually very close to the peg, but its risk is concentrated in rare, extreme episodes; (c) the data contain an error; (d) USDC deviations are normally distributed.

13. A Curve StableSwap pool with amplification $A = 2000$ quotes USDC at 0.997 while 90% of the pool is USDC. This indicates (a) USDC is safe, since the price is close to \$1; (b) a balanced pool; (c) the pool is becoming one-sided, a sign of stress the price does not yet show; (d) a data error.

14. A price-only alarm for stablecoin depegs has AUC 0.90 on test hours but detects none of the 8 test events in time. The explanation is (a) AUC is always misleading; (b) the data are wrong; (c) 8 events are too many; (d) a high AUC comes from hours inside episodes, while an alarm must fire before an episode starts.

15. In the stablecoin depeg pressure index (DPI) a higher average selected Quantile Lasso penalty $\hat{\lambda}$ indicates (a) stronger co-movement in the lower tail, with fewer but stronger links; (b) more active predictors; (c) lower volatility; (d) a better forecast.

16. The Quantile Lasso equations behind the DPI fit the lower tail of stablecoin deviations well in sample, yet held-out forecast skill is negative. The lesson is (a) the model is wrong; (b) in-sample fit is not out-of-sample forecasting ability; (c) more data always help; (d) the tail is normal.

17. In a Diebold–Yilmaz spillover table, $\mathrm{NET}_{\mathrm{DOGE}} = -18.6$. This means (a) Dogecoin fell 18.6%; (b) Dogecoin causes the other coins; (c) Dogecoin receives more forecast-error variance from the others than it transmits; (d) Dogecoin is uncorrelated with the market.

18. The Merge has $\mathrm{CAR}[-1, 1] = -11.1\%$ ($t = -3.09$); across 13 Ethereum upgrades the mean CAR is -0.18% ($p = 0.92$). The correct conclusion is (a) single upgrades can move the price strongly, but there is no common revaluation; (b) upgrades always destroy value; (c) the Merge result is an error; (d) all upgrades are significant.

19. XGBoost classifies tomorrow's Bitcoin direction with 78.3% accuracy in sample and 51.3% out of sample; 50.6% of days are up days. This is (a) strong evidence of predictability; (b) overfitting: the model learnt noise; (c) a data error; (d) proof of market inefficiency.

20. The best of 810 trading rules has Sharpe ratio 0.84 in 2018–2022; its deflated Sharpe ratio is 0.72. Hence (a) the rule is significant at 1%; (b) the rule will earn 0.84 in the future; (c) the Sharpe ratio is wrong; (d) once the number of trials is counted, the rule is not significant.

Case 1: Bitcoin Volatility and VaR Backtest

GARCH(1,1)-t	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	half-life	$\hat{\nu}$
Bitcoin	0.144 (0.058)	0.076 (0.012)	0.924 (0.012)	1.000	∞	3.13 (0.15)
S&P 500	0.035 (0.009)	0.162 (0.022)	0.819 (0.021)	0.981	36	5.85 (0.71)

1-day VaR 1%, Bitcoin, 2020–2026	hits	rate (%)	Kupiec p	indep. p	last 250	Basel zone
Normal, 250 days	45	1.83	< 0.001	0.850	5	amber
HS, 500 days	25	1.02	0.924	0.256	3	green
EWMA, Normal	51	2.08	< 0.001	0.952	5	amber
GARCH(1,1)-t	28	1.14	0.491	0.329	2	green

Daily log returns in %; robust standard errors in brackets; 2 453 test days, 24.5 hits expected. Mean VaR: HS 8.64%, GARCH-t 8.12%. Kupiec p: test of a 1% breach rate; indep. p: test that breaches do not cluster.

21. The degrees of freedom $\hat{\nu}$ are 3.13 for Bitcoin and 5.85 for the S&P 500. This shows that (a) Bitcoin volatility is less persistent than S&P 500 volatility; (b) Bitcoin returns have about half the variance of S&P 500 returns; (c) standardised Bitcoin shocks stay heavy-tailed after GARCH filtering; (d) GARCH filtering removes the heavy tails of both return series completely.

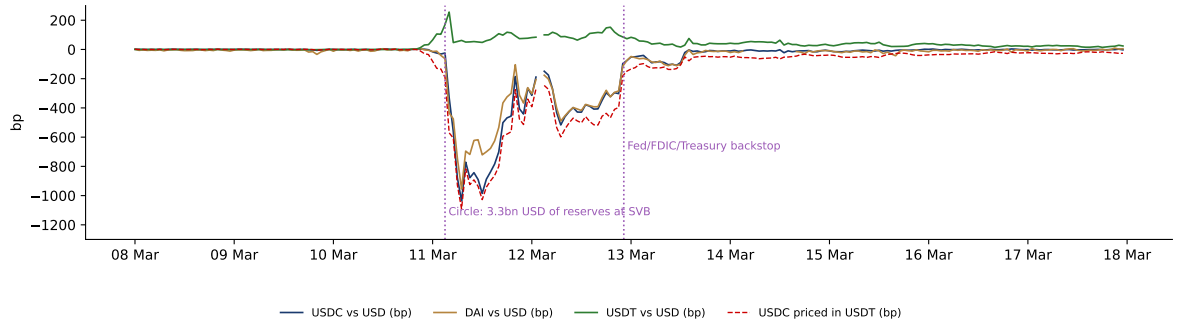
22. Which caveat applies to the Bitcoin persistence estimate $\hat{\alpha}+\hat{\beta}=1.000$? (a) it can also reflect structural breaks between very different market phases; (b) it proves that Bitcoin volatility never returns to a long-run mean; (c) it is invalid because robust standard errors were used in estimation; (d) it implies a half-life of 36 trading days, exactly as for the S&P 500 index.

23. The Normal and EWMA-Normal VaR have Kupiec $p < 0.001$ but independence p of 0.850 and 0.952. The justified reading is (a) their breaches cluster in time, so the volatility filter reacts too slowly; (b) they hold too much capital, since the VaR is breached far too rarely in the sample; (c) they pass the backtest, since the independence test is not rejected; (d) they breach too often but not in clusters: the Normal quantile is too thin.

24. A risk manager must choose between HS and GARCH-t, both in the green zone. Which argument favours GARCH-t? (a) its breach rate of 1.14% is closer to the promised 1% than that of HS; (b) it gives correct coverage with a lower mean VaR (8.12% vs 8.64%); (c) it has more breaches in the last 250 days than HS; (d) it is the only model whose Kupiec test is not rejected.

25. Which further evidence would best separate HS from GARCH-t? (a) a longer list of breach rates computed on exactly the same test days; (b) a Jarque–Bera test of normality on the raw Bitcoin returns; (c) quantile scores with a DM test, plus ES backtests by sub-period; (d) the number of breaches in the last 250 days of the sample.

Case 2: A Stablecoin under Stress



Hourly deviations from \$1 in bp around the Silicon Valley Bank (SVB) failure, March 2023. Circle held 3.3 bn USD of about 40 bn USD reserves at SVB (8.25%). Minimum: USDC −1 036 bp and DAI −955 bp at 07:00 UTC on Saturday 11 March. Early-warning model of the lecture (paper): 72% of test events detected, on average 14 hours ahead, with 2.5 false alerts a month; price-only alarms meet the false-alert budget only by detecting almost no events.

26. If Circle had lost its entire SVB deposit (8.25% of reserves), the fair value of one USDC would deviate from \$1 by (a) −330 bp; (b) −825 bp; (c) −1 036 bp; (d) −211 bp.

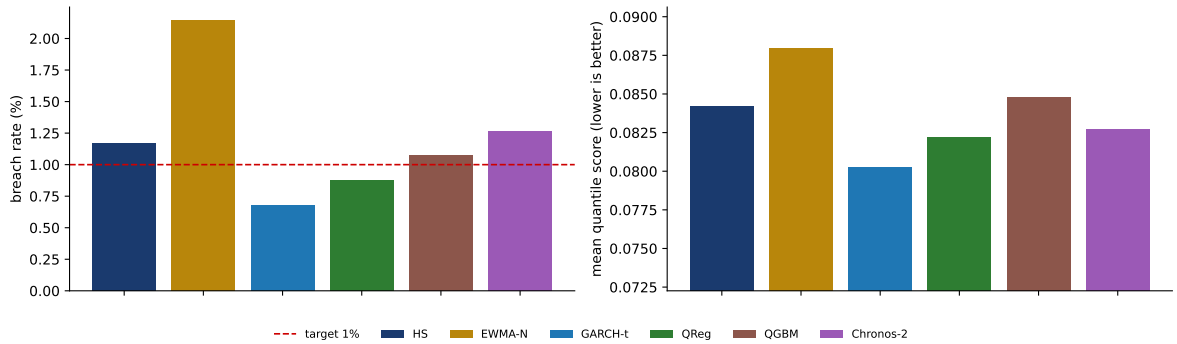
27. USDC traded down to −1 036 bp while about 92% of its reserves were untouched. The episode shows that (a) the remaining reserves were too small to back the token; (b) the market priced exactly the expected credit loss on SVB; (c) a fully backed stablecoin cannot deviate from \$1 for more than a few hours; (d) a peg can fail when redemption is closed, even with most backing intact.

28. DAI fell to −955 bp at the same hour, while USDT rose above \$1. The best explanation is (a) much of DAI's collateral was USDC, and holders fled to the other dollar coin; (b) DAI held its own deposits at SVB, while USDT was backed by the Federal Reserve; (c) DAI and USDT are both issued by Circle, so they moved in opposite ways; (d) the DAI move is a data error, since a decentralised coin cannot depeg.

29. The early-warning model detects 72% of test events, on average 14 hours ahead, with 2.5 false alerts a month. The correct reading is (a) 72% of all test hours were classified correctly by the model; (b) each alert is followed by a depeg within the next 14 hours; (c) 72% of test episodes were flagged before they began; (d) the model will detect exactly 72% of future depeg episodes.

30. Why monitor the composition of a StableSwap pool and not only its price, and what is the limitation? (a) the price moves first, but composition is cheaper to observe on chain; (b) composition moves first, but it also shifts for benign reasons (incentives); (c) composition and price move together, so either one is sufficient; (d) composition moves first, and any shift in it signals a coming depeg with certainty.

Case 3: Machine Learning and Risk Forecasts



model	breach %	Kupiec p	QS	DM p*
HS	1.17	0.60	0.0842	0.45
EWMA-N	2.14	0.001	0.0880	0.18
GARCH-t	0.68	0.28	0.0803	–
QGBM (boosted trees)	1.07	0.82	0.0848	0.36
Chronos-2 (pretrained)	1.27	0.41	0.0827	0.54

1-day VaR 1% for Bitcoin, January 2024 – October 2026: 1 027 test days, 10.3 breaches expected. *DM p: test of equal mean QS against GARCH-t. QS: mean quantile score, lower is better. QGBM: quantile gradient boosting machine. Chronos-2 was released in October 2025.

31. GARCH-t has the lowest QS (0.0803) although only 0.68% of days are breaches. This is because (a) a lower QS means more breaches, so GARCH-t is in fact too aggressive; (b) the QS rewards the model whose breach rate is closest to 1%; (c) GARCH-t fails the Kupiec test, so its QS cannot be used; (d) the QS judges daily closeness to the quantile, not the breach count.

32. All DM p-values against GARCH-t are 0.18 or higher. The justified conclusion is (a) we do not reject equal mean QS, so the ranking is not conclusive; (b) GARCH-t is significantly better than every other model; (c) all five models give the same forecast on every day; (d) HS is significantly worse than GARCH-t at the 45% level.

33. EWMA-N fails the coverage test (Kupiec $p = 0.001$), yet its QS is not significantly worse than that of GARCH-t (DM $p = 0.18$). Both can hold because (a) the Kupiec test is invalid for models with Normal quantiles; (b) Kupiec counts breaches, while the DM test on QS has little power here; (c) the DM test checks breach clustering, not the breach count; (d) EWMA-N has too few breaches, which the quantile score does not penalise.

34. Should EWMA-N be used for a 1% VaR? (a) yes, since its QS is not significantly worse than that of the GARCH-t model; (b) no: its volatility filter should be replaced by a constant variance; (c) no: failed coverage is a problem in itself, and the fix is the distribution; (d) yes, as long as the test period is extended to more than 2 000 days.

35. Chronos-2 performs as well as GARCH-t without any fitting to Bitcoin. The main concern about this comparison is (a) its QS of 0.0827 is significantly worse than that of GARCH-t; (b) a pretrained model cannot produce quantile forecasts at all; (c) it was not fitted to Bitcoin, so it cannot contain any look-ahead bias; (d) its training data may include Bitcoin prices from the test period.